

Overdrive-01-KoT

Project Report

1. Readme

KoT

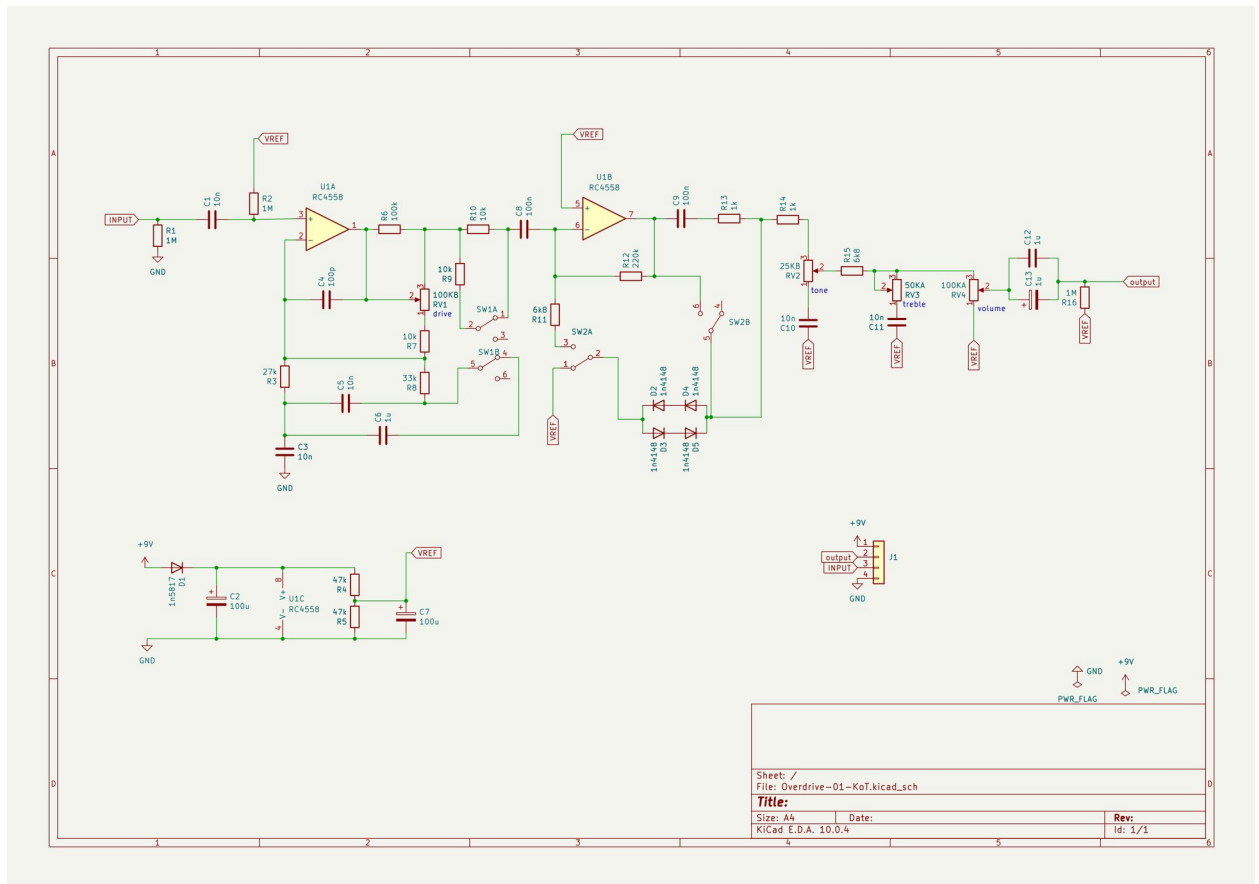
This is an overdrive pedal modeled after the King Of Tone pedal.

Version 1 had error making the drive pot useless. V1 was bodged by putting a link between pad2 of Rv1 (drive) to the 100k R6, and which is connected to pin1 of U1. V3 removed the via from v2, and reversed the drive pot to match the schematic.

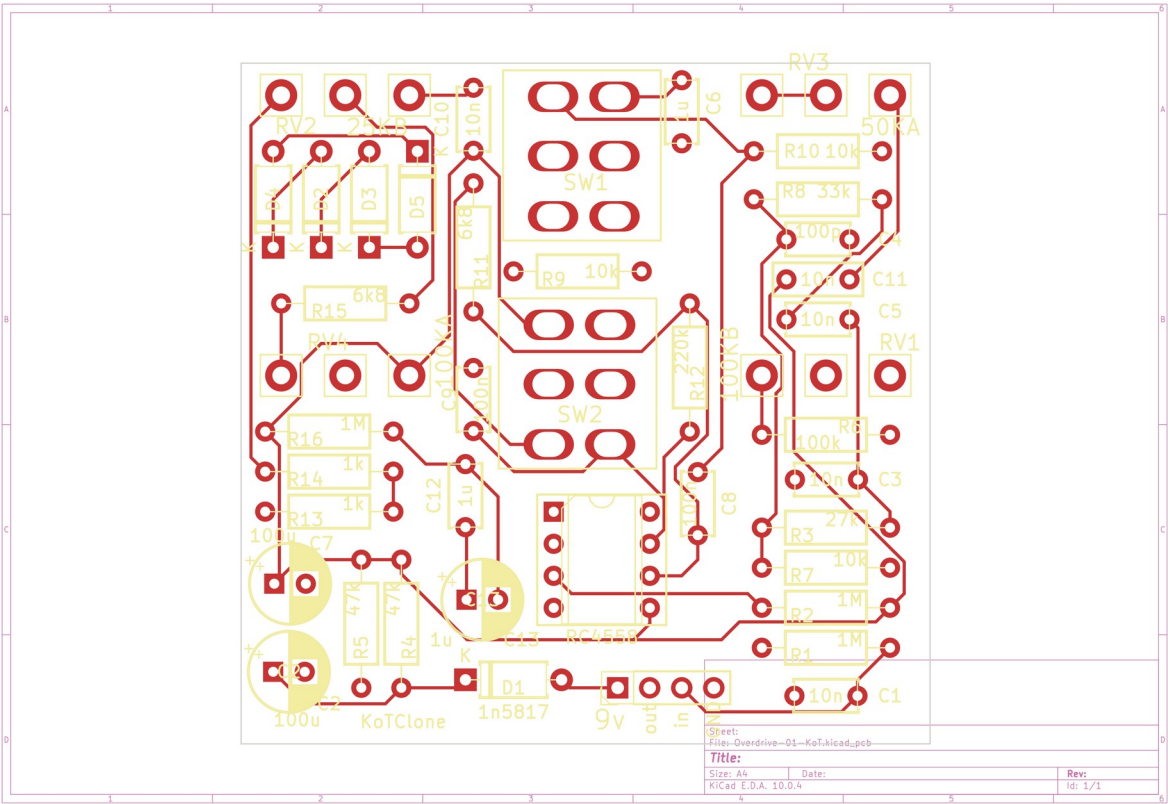
The current build seems to be fully functional.

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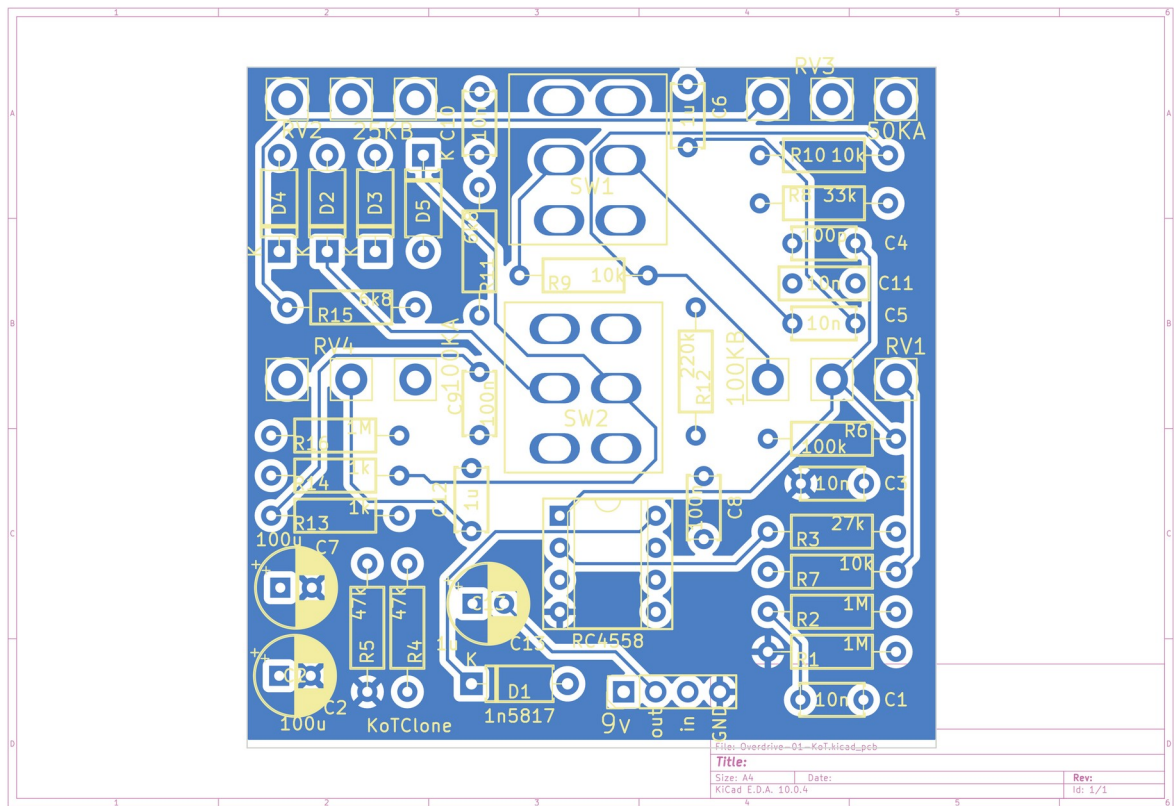
2. Schematic



3. PCB F.Cu + Silkscreen



4. PCB B.CU + Silkscreen



5. Bill of Materials

Reference	Value	Footprint	Qty	DNP
C1,C3,C5,C10	10n	Capacitor_THT: C_Disc_D5.0mm_W2.5mm_P5.00mm	4	
C2,C7	100u	Capacitor_THT: CP_Radial_D6.3mm_P2.50mm	2	
C4	100p	Capacitor_THT: C_Disc_D5.0mm_W2.5mm_P5.00mm	1	
C6,C12	1u	Capacitor_THT: C_Disc_D5.0mm_W2.5mm_P5.00mm	2	
C8,C9	100n	Capacitor_THT: C_Disc_D5.0mm_W2.5mm_P5.00mm	2	
C11	10n	Capacitor_THT: C_Rect_L7.0mm_W2.5mm_P5.00mm	1	
C13	1u	Capacitor_THT: CP_Radial_D6.3mm_P2.50mm	1	
D1	1n5817	Diode_THT:D_A-405_P7.62mm_Horizontal	1	
D2-D5	1n4148	Diode_THT:D_A-405_P7.62mm_Horizontal	4	
J1	Conn_01x04	Connector_PinHeader_2.54mm:PinHeader_	1	

		1x04_P2.54mm_Vertical		
R1,R2,R16	1M	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizontal	3	
R3	27k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizontal	1	
R4,R5	47k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizontal	2	
R6	100k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizontal	1	
R7,R9,R10	10k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizontal	3	
R8	33k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizontal	1	
R11,R15	6k8	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1	2	

		6mm_Horizon tal		
R12	220k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	1	
R13,R14	1k	Resistor_THT: R_Axial_DIN02 07_L6.3mm_D 2.5mm_P10.1 6mm_Horizon tal	2	
RV1	100KB	pedals:16MM_ PCB_PADS	1	
RV2	25KB	pedals:16MM_ PCB_PADS	1	
RV3	50KA	pedals:16MM_ PCB_PADS	1	
RV4	100KA	pedals:16MM_ PCB_PADS	1	
SW1	SW_DPST_x2	pedals:DPDT Mini	1	
SW2	SW_DPDT_x2	pedals:DPDT Mini	1	
U1	RC4558	Package_DIP: DIP- 8_W7.62mm_S ocket	1	

Total line items: 26